

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Sixth Semester of B. Tech. (IT) Examination****May 2018****IT319 Capstone Course****Date: 05.05.2018, Saturday****Time: 10:00 a.m. To 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer following questions.****[07]**

1. What is worst case complexity of Quick sort and Merge sort?
2. The height of a binary tree is the maximum number of edges in any root to leaf path.
The maximum number of nodes in a binary tree of height h is:
3. How many stacks are needed to implement a queue? Consider the situation where no other data structure like arrays, linked list is available to you.
4. Let T be a tree with 10 vertices. The sum of the degrees of all the vertices in T is _____.
5. Define Max-Heap Tree.
6. Four jobs to be executed on a single processor system arrive at time 0 in the order A, B, C, D. their burst CPU time requirements are 4, 1, 8, 1 time units respectively. The completion time of A under round robin scheduling with time slice of one time unit is (a) 10 (b) 4 (c) 8 (d) 9
7. Assume that for a certain processor, a read request takes 50 nanoseconds on a cache miss and 5 nanoseconds on a cache hit. Suppose while running a program, it was observed that 80% of the processor's read requests result in a cache hit. The average read access time in nanoseconds is _____.

Q - 2(a) Construct binary search tree for following data.**[05]**

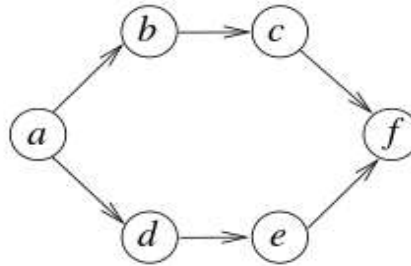
99, 29, 79, 49, 69, 59, 62, 60, 64, 63

After creation of BST apply post-order traversal.

Q - 2(b) Attempt the following. (ANY THREE)**[09]**

1. A uniprocessor computer system only has two processes, both of which alternate 10 ms CPU bursts with 90 ms I/O bursts. Both the processes were created at nearly the same time. The I/O of both processes can proceed in parallel. Which of the scheduling strategies will result in the least CPU utilization (over a long period of time) for this system? Justify.

2. Discuss different applications of stack data structure.
3. Explain Implications of dangling pointer in detail.
4. Consider the following directed graph. Write total number of topological orderings for the given graph.



List out all topological orderings.

5. What is pointer to function? How it differs from pointer to variable?

Q - 3 (a) Consider a storage disk with 4 platters (numbered as 0, 1, 2 and 3), 200 cylinders (numbered as 0, 1, ... , 199), and 256 sectors per track (numbered as 0, 1, ... , 255). The following 6 disk requests of the form [sector number, cylinder number, platter number] are received by the disk controller at the same time: **[04]**

[120, 72, 2] , [180, 134, 1] , [60, 20, 0] , [212, 86, 3] , [56, 116, 2] , [118, 16, 1]

Currently the head is positioned at sector number 100 of cylinder 80, and is moving towards higher cylinder numbers. The average power dissipation in moving the head over 100 cylinders is 20 mill watts and for reversing the direction of the head movement once is 15 mill watts. Power dissipation associated with rotational latency and switching of head between different platters is negligible.

The total power consumption in mill watts to satisfy all of the above disk requests using the Shortest Seek Time First disk scheduling algorithm is _____.

Q - 3 (b) Attempt the following. (ANY TWO) **[10]**

1. List the major I/O scheduling algorithm. Explain any one in detail
2. Explain pointer to array in detail with suitable example.
3. Why do we need postfix expression? Convert the given infix expression into postfix expression using Stack.

$$(A + B * C) - (D * E) * F / G$$

SECTION – II**Q – 4 Do as directed. [07]**

1. To construct 7 x 128 decoder how many 2 x 4 decoder are required? 01
2. Select appropriate option. 01

Which of the following scenarios may lead to an irrecoverable error in a database system?

- (A) A transaction writes a data item after it is read by an uncommitted transaction
 - (B) A transaction reads a data item after it is read by an uncommitted transaction
 - (C) A transaction reads a data item after it is written by a committed transaction
 - (D) A transaction reads a data item after it is written by an uncommitted transaction
3. Select appropriate option for following C code. 01

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int arri[] = {1, 2, 3};
```

```
    int *ptri = arri;
```

```
    char arrc[] = {1, 2, 3};
```

```
    char *ptrc = arrc;
```

```
    printf("%d ", sizeof(arri));
```

```
    printf("%d ", sizeof(ptri));
```

```
    printf("%d ", sizeof(arrc));
```

```
    printf("%d ", sizeof(ptrc));
```

```
    return 0;
```

```
}
```

(A) 3 4 3 4

(C) 3 4 3 1

(B) 12 4 3 1

(D) 12 4 3 4

4. What is 2's complement representation of $(23)_{10}$? 02
5. Select appropriate option for following question. 02

Consider a computer with a 4-ways set-associative mapped cache of the following characteristics: a total of 1 MB of main memory, a word size of 1 byte, a block size of 128 words and a cache size of 8 KB. The number of bits in the TAG, SET and WORD fields, respectively are: _____

(A) 9,5,7

(C) 8,5,7

(B) 8,6,6

(D) 9,4,7

Q – 5 (a) Attempt the following. [07]

1. Select appropriate option for following code. [02]
Consider the following program written in pseudo-code. Assume that x and y are integers.

```
Count(x, y) {
```

```
    if (y != 1) {
```

```
        if (x != 1) {
```

```
            print("*");
```

```
            Count (x/2, y);
```

```
        }
```

```

    else {
        y=y-1;
        Count (1024, y);
    }
}
}

```

The number of times that the print statement is executed by the call Count(1024, 1024) is _____.

- (A) 10230 (C) 1023
(B) 10 (D) 23010

2. Consider the following C program:

[02]

```

#include <stdio.h>
void fun1(char *s1, char *s2)
{
    char *temp;
    temp = s1;
    s1 = s2;
    s2 = temp;
}
void fun2(char **s1, char **s2)
{
    char *temp;
    temp = *s1;
    *s1 = *s2;
    *s2 = temp;
}

int main()
{
    char *str1 = "Hi", *str2 = "Bye";
    fun1(str1, str2);
    printf("%s %s", str1, str2);
    fun2(&str1, &str2);
    printf("%s %s", str1, str2);
    return 0;
}

```

The output of the program above is

- (A) Hi Bye Bye Hi (B) Hi Bye Hi Bye (C) Bye Hi Hi Bye (D) Bye Hi Bye Hi

3. Draw precedence graph for the following schedule consisting of four transactions and determine whether it is conflict Serializable OR not. If so give the serial sequence for Serializability. [03]

T1	T2	T3	T4
read(X)			
	read(X)		
write(Y)			
		read(Y)	
	read(Y)		
	write(X)		
		read(W)	
		write(Y)	
			read(W)
			read(Z)
			write(W)
read(Z)			
write(Z)			

Q - 5 (b) Consider the given relation and functional dependencies: [03]

$R(A, B, C)$

$FD = \{ AB \rightarrow C, C \rightarrow A \}$

The above relation is in which normal form? Justify.

OR

Q - 5 (b) Every candidate key can be super key but there are many super keys which are not candidate keys. Is that true? Justify. [03]

Q - 5 (c) Given $R(A, B, C, D, E)$ with functional dependency as: $A \rightarrow C$, $AB \rightarrow D$ and $D \rightarrow E$. Now relation is decomposed into $R_1(A, C)$, $R_2(A, B, D)$ and $R_3(D, E)$. Check If the given decomposition is lossless join decomposition or not. [04]

OR

Q - 5 (c) Compute the canonical cover for the below example. [04]
 $R(A, B, C, D)$ with functional dependency $F = \{ A \rightarrow BC, B \rightarrow C, AB \rightarrow D \}$.

Q - 6 (a) Attempt the following. [05]

1. Some Nonstandard floating representation is used by a machine. Here S is for sign bit, E is for exponent and M is for mantissa. If the machine uses bias-K, what will be the value of K? [02]

S(1)	E(7)	M(24)
------	------	-------

2. To construct 7 x 128 decoder how many 2 x 4 decoder are required? Also draw block diagram of 2 x 4 decoder. [03]

Q - 6 (b) Attempt Following. (ANY THREE) [09]

1. Every view serializable schedule is not conflict Serializable. Justify.
2. Differentiate calloc and malloc function. By how many ways a previously allocated memory can be deallocate in C.
3. Given $f(x, y, w, z) = \sum m(0, 1, 2, 3, 7, 8, 10) + \sum d(5, 6, 11, 15)$, where d represents the don't-care condition in Karnaugh maps. Which of the following is a minimum sum of products (SOP) form of $f(x, y, w, z)$?
4. A memory organization has two level cache organization. Access time for L1, L2 and M is 5ns, 50ns and 800ns respectively. Miss rate for L1 and L2 is 0.13 and 0.08 respectively. What is average memory access time for the same?